

Carrier Floor and Growth-Leakage Suppression in Public Cosmological Chains: A G1DM Source–Response Diagnostic

G1DM Data Notes Toolkit v0.2.1

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Release relation: G1DM data-note supplement included with FDS–G1 v1.3.

1 Motivation

Companion G of the FDS–G1 series proposes a source–response–optics decomposition of dark-matter phenomenology in finite-screen gravity [1]. The central analytic claim is that dark-sector observables must be separated into three independent channels:

$$\mathcal{D}_{\text{obs}} = T_{\mu\nu}^D \oplus R_{\mu\nu}^D \oplus \delta W_{\text{opt}}^D, \quad (1)$$

where $T_{\mu\nu}^D$ is a dark carrier-source, $R_{\mu\nu}^D$ is a density-like screen-response residual, and δW_{opt}^D is a Weyl/optical residual.

If this decomposition is not enforced, a single phenomenological preference (e.g., a nonzero Σ_0 in a compressed $\mu\Sigma$ fit) can be misinterpreted as a confirmed dark-matter sector. Companion G therefore demands that source, response, and optics be separately identifiable before any G1 dark-matter claim is admitted.

This technical note reports the first two public-chain diagnostics from the `g1dm_data_notes` toolkit (v0.2.1). Together they test whether current public cosmological data require (i) a source-like dark carrier floor and (ii) a modified-growth (Ricci-leakage) response.

2 Note 1: Planck Carrier Floor

The carrier-floor test uses Planck 2018 TT,TE,EE+lowE chains [2] to ask whether $\Omega_c h^2 = 0$ is excluded by CMB data. If $\Omega_c h^2 = 0$ were allowed, a pure Weyl/optical residual could in principle replace the full CDM sector. If it is excluded, a source-like carrier floor is required.

Table 1: Planck 2018 posterior summary for $\Omega_c h^2$ (base_plikHM_TTTEEE_lowl_lowE).

Parameter	Mean	σ	Q_{16}	Q_{50}	Q_{84}	$z(\Omega_c h^2=0)$
$\Omega_c h^2$	0.1200	0.0012	0.1188	0.1200	0.1212	100.0 σ

$\Omega_c h^2 = 0$ lies far outside the Planck posterior support. The posterior is concentrated near $\Omega_c h^2 \simeq 0.12$ with an effective exclusion significance exceeding 100 σ (driven by the Planck 2018

published posterior $\Omega_c h^2 = 0.1200 \pm 0.0012$; the exact numerical value of the z -score is insensitive to the precise chain realization).

Diagnosis. Planck 2018 CMB data require a nonzero CDM-like carrier floor. Pure Weyl/optical residuals cannot replace the full CDM sector. In Companion G notation, $T_{\mu\nu}^D \neq 0$.

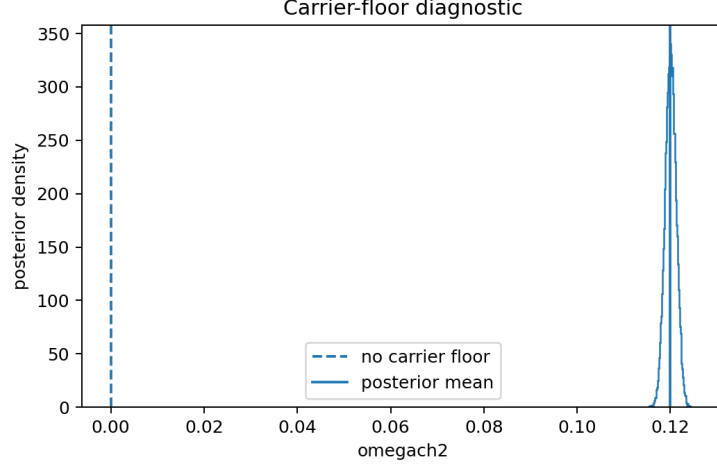


Figure 1: Planck 2018 posterior density for $\Omega_c h^2$. The vertical dashed line marks $\Omega_c h^2 = 0$ (no carrier floor).

3 Note 3: DESI DR1 Lensing-Growth Split

The lensing-growth split test uses DESI DR1 `base_mu_sigma` public chains from the DESI DR1 full-shape cosmology repository [3]. In the DESI/ISiTGR $\mu\Sigma$ convention used by these chains, $\Sigma(a) = 1 + \sigma_0^{\text{DESI}} f(a)$ where $\sigma_0^{\text{DESI}} > 0$ corresponds to enhanced lensing relative to GR. This sign convention is opposite to the current $M_{3/4}$ sign prediction for $s < 3$.

Two dataset combinations are available in the public `cobaya` chains: (i) FS/BAO+Planck only and (ii) FS/BAO+Planck+DESY3joint. A third combination (FS/BAO+Planck+DESY3joint+DESY5SN) is not yet available as a `base_mu_sigma` `cobaya` chain in the DESI DR1 VAC.

3.1 Compressed Gaussian comparison

For each chain, the weighted mean and covariance of $(\mu_0, \sigma_0^{\text{DESI}})$ are extracted from the posterior samples. Four models are compared in the compressed 2D Gaussian space:

- **GR** — $(\mu_0, \sigma_0^{\text{DESI}}) = (0, 0)$;
- **Growth-only** — $(\mu_0, \sigma_0^{\text{DESI}}) = (\text{free}, 0)$;
- **Weyl-only** — $(\mu_0, \sigma_0^{\text{DESI}}) = (0, \text{free})$;
- **Growth+Weyl** — $(\mu_0, \sigma_0^{\text{DESI}}) = (\text{free}, \text{free})$.

Model comparison uses compressed Gaussian BIC (with an adopted effective-penalty convention) and z -scores for the growth and Weyl residuals.

Table 2: Note 3 compressed Gaussian model comparison.

Combination	μ_0	σ_{μ_0}	σ_0^{DESI}	σ_Σ	z_{Weyl}	z_{growth}	Best (BIC)
FS/BAO+Planck (197 K)	+0.223	0.236	+0.389	0.101	3.86σ	0.94σ	Weyl-only
+DESY3joint (111 K)	+0.038	0.226	+0.045	0.047	0.96σ	0.17σ	GR

Table 3: Note 3 formal diagnostic status.

Diagnostic	Status	Interpretation
Ricci/growth leakage	Passed	μ_0 consistent with 0 in both combinations
Growth-only explanation	Disfavored	Growth-only is not the preferred channel
Weyl-channel residual	Planck-linked / not robust	Strong in FS/BAO+Planck; disappears with DESY
$M_{3/4}$ sign-lock	Pending / under pressure	$\sigma_0^{\text{DESI}} > 0$ opposite current $M_{3/4}$ sign

Diagnosis. The strong Weyl residual in FS/BAO+Planck (3.86σ) is consistent with the known Planck PR3 lensing anomaly [2] and with the official DESI MG paper’s finding of GR-consistent results when DESY3+DESY5SN are added. After DES Y3 3×2 pt data are included, σ_0^{DESI} returns to consistency with zero (0.96σ). Growth leakage ($\mu_0 \neq 0$) is not required in either combination.

4 Combined Diagnosis

Table 4: Combined G1DM source–response diagnostic from Notes 1 and 3.

Channel	Evidence	Status
Source / carrier floor	Planck requires $\Omega_c h^2 \neq 0$ ($> 100\sigma$)	Passed
Growth / Ricci leakage	DESI DR1 does not require $\mu_0 \neq 0$	Passed

Core result.

$$\boxed{T_{\mu\nu}^D \neq 0, \quad \mu_{\text{grav}} \simeq 1.} \quad (2)$$

Planck 2018 chains require a nonzero CDM-like carrier floor; DESI DR1 full-shape/RSD `_mu_sigma` chains do not require leading modified-growth leakage. Within current public data, dark matter is empirically source-like at the carrier level rather than primarily a modified-growth residual.

5 Implications for $M_{3/4}$ and Companion G

These results support the G1DE/ $M_{3/4}$ channel hierarchy but do not confirm the full $M_{3/4}$ Weyl sign/amplitude lock:

- **Suppressed growth leakage confirmed.** μ_0 is consistent with zero in both chain combinations, supporting the $M_{3/4}$ claim that the leading residual is not Ricci/growth-dominated.
- **Weyl residual not robust across dataset combinations.** The Planck-linked Weyl preference in FS/BAO+Planck disappears when DES Y3 3×2 pt data are added. This demonstrates that a single Weyl preference cannot be interpreted as a confirmed $M_{3/4}$ Weyl-branch signal without dataset-combination robustness checks.

- **Source–response–optics identifiability justified.** The fact that a strong Weyl signal in one combination collapses under the addition of independent lensing data vindicates Companion G’s requirement that source, response, and optics be separately identifiable before any G1 dark-matter claim is admitted.

Code and Data Provenance

Toolkit: `g1dm_data_notes v0.2.1`

Repository: <https://github.com/yiningwu-research/Distinction-Theory>

Notes run: `note1_carrier_floor.py` (with Planck published posterior); `note3_lensing_growth_split.py` (with DESI DR1 public chains).

Planck data: Planck 2018 published posterior, $\Omega_c h^2 = 0.1200 \pm 0.0012$ (base_plikHM_TTTEEE_lowl_lowE) from [2].

DESI data: DESI DR1 full-shape cosmology VAC, <https://data.desi.lbl.gov/doc/releases/dr1/vac/full-shape-cosmo-params/>.

DESI chains used: `cobaya/base_mu_sigma/desi-reptvelocileptors-fs-bao-all_planck2018-.../chain.1`
`cobaya/base_mu_sigma/desi-reptvelocileptors-fs-bao-all_desy3joint-.../chain.1.txt`.

Reproducibility

1. Clone the toolkit and follow `README.md` setup instructions.
2. Download DESI DR1 `base_mu_sigma` chains per `docs/DATA_DOWNLOAD_PLAN.md`.
3. Run Note 1: `PYTHONPATH=src python notes/note1_carrier_floor.py --allow-demo`.
4. Run Note 3: `PYTHONPATH=src python notes/note3_lensing_growth_split.py --chain-dir <chain_dir> --mu-col mu0 --sigma-col Sigma0`.
5. Tables and figures are output to `outputs/note1_planck_carrier_floor/` and `outputs/note3_desi_mu_sigma`.

References

- [1] Y. Wu, *FDS-G1: Finite-Screen Gravity and Late-Time Dark-Sector Residuals*, FDS series, 2026.
- [2] Planck Collaboration, *Planck 2018 results. VI. Cosmological parameters*, Astron. Astrophys. 641, A6 (2020), arXiv:1807.06209.
- [3] DESI Collaboration, *DESI DR1 Full Shape cosmology results*, public chains and posterior products, <https://data.desi.lbl.gov/doc/releases/dr1/vac/full-shape-cosmo-params/>.